

<page_start no = 1>

<sol_start id=1>

1. Convex lens is used to produce real image.

<sol_end>

<sol_start id=2>

2.
$$\text{Power (P)} = \frac{1}{\text{Focallength(f) in m}}$$

In case I :

'f = 10 cm given

$$f = \frac{10}{100} = 0.1 \text{ m}$$

$$p = \frac{1}{0.1} = 10\text{D}$$

In case II

f = 20 cm

$$f = \frac{20}{100} = 0.2$$

$$p = \frac{1}{0.2} = 5\text{D}$$

<sol_end>

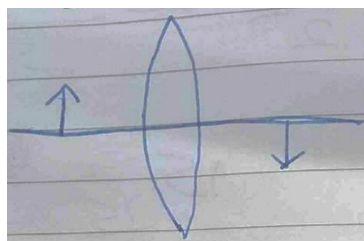
<page_end>

<page_start no = 2>

<sol_start id=3>

3. u = -5 cm

v = +10 cm



$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{10} + \frac{1}{5}$$

$$\frac{1}{f} = \frac{1+2}{10} = \frac{3}{10}$$

$$f = \frac{10}{3} = 3.33$$

∴ It is convex lens because f is +ve

<sol_end>

<sol_start id=4>

4. $u = -50 \text{ cm}$

$v = 50 \text{ cm}$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{50} - \left(\frac{1}{-50} \right)$$

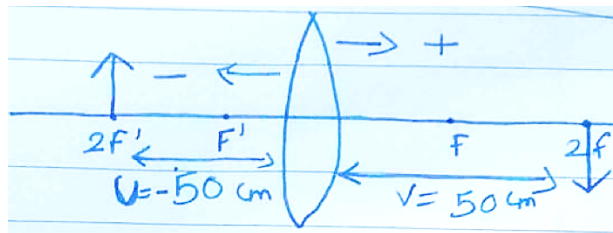
$$\frac{1}{f} = \frac{1}{50} + \frac{1}{50}$$

$$\frac{1}{f} = \frac{2}{50}$$

$$\frac{1}{f} = \frac{1}{25}$$

$f = +25 \text{ cm}$

$$f = \frac{25}{100} \text{ m}$$



$$f = 0.25 \text{ m}$$

$$\text{Power} = \frac{1}{\text{focallength}}$$

$$p = \frac{1}{f}$$

$$p = \frac{1}{0.25} = +4\text{D}$$

<sol_end>

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